

Co-Evolutionary Genetic Algorithms for Memorization-Constrained Robust Chess Opening Repertoires

Minhaj ul Hassan, Zainab Zahid, and Hazeera Muhammad Hashim

Abstract—We study whether diversity-preserving co-evolutionary adversarial training improves the robustness of memorization-constrained chess opening repertoires. A repertoire commits to at most 25 moves per color and must cover any opponent reply played in $\geq 15\%$ of games at a reached position (the *closure rule*). We compare four methods across 15 independent seeds: a greedy most-played-move heuristic, a static GA trained against a uniform rating-band mixture (STATIC), a greedy hill-climber (GREEDY_HILLCLIMB) with equal evaluation budget, and a co-evolved opponent population with explicit diversity preservation (COEVOLVE).

All GA variants significantly outperform the most-played-move heuristic (held-out mean 0.526 vs. 0.494, $p < 0.001$, $A_{12} = 0.00$). Unlike naïve co-evolution, the diversity-preserving implementation keeps opponent behavioral diversity stable throughout 100 generations (mean pairwise ℓ_2 distance ≈ 0.52 , flat from generation 0 to 100). Under this regime, COEVOLVE shows a positive trend over STATIC (mean 0.5260 vs. 0.5255, $A_{12} = 0.60$) and the full population-based GA beats the hill-climber (STATIC vs. GREEDY: $A_{12} = 0.70$). However, neither gap reaches statistical significance at $\alpha = 0.05$ (COEVOLVE vs. STATIC: $p = 0.49$). We trace the persistent null result to train-eval distributional symmetry: both training fitness and held-out evaluation reduce to expected score under the same uniform rating-band mixture, leaving co-evolution no distributional shift to exploit. A closure-rule ablation confirms the novel structural constraint is load-bearing (removal costs ≈ 0.008 held-out score, $p < 0.001$).

Index Terms—Chess opening repertoire, genetic algorithm, co-evolutionary optimization, robust optimization, CVaR, diversity preservation, constrained combinatorial optimization.

I. INTRODUCTION

CHES opening preparation is fundamentally a constrained memorization problem. A club-level player can realistically commit a few dozen specific positions to long-term memory, yet the branching tree of plausible opponent replies within the first five moves alone runs into thousands of positions. This tension—finite memory, exponential branching—makes opening preparation an instance of constrained combinatorial optimization under uncertainty.

The standard human approach is deterministic and fragile: memorize the moves most often played at your rating level. This heuristic ignores two structural problems. First, a move that is popular on average may perform poorly against specific sub-populations of opponents (e.g., players in a particular rating band who favor unusual lines). Second, it treats every

opponent reply as equally likely, even rare ones that can be catastrophic if encountered unprepared.

We treat the problem as *robust constrained optimization*: build a repertoire that performs well not just in expectation but against adversarial opponent mixtures drawn from a realistic distribution. The key mechanisms are: (i) a memorization budget of at most 25 committed moves per color; (ii) a *closure rule* that automatically covers any opponent reply played in $\geq 15\%$ of games; and (iii) a CVaR-based fitness function that penalizes worst-case performance across rating bands.

The central design question is *what opponent model to train against*. A static uniform mixture is simple and unbiased but may miss adversarial sub-distributions. Co-evolutionary adversarial training—evolving a population of opponents alongside repertoire candidates—promises to discover the candidate’s worst case dynamically [1], [2]. We ask:

Under what conditions does diversity-preserving co-evolutionary adversarial training improve generalization for memorization-constrained chess repertoire optimization?

We implement the full pipeline including explicit diversity-preservation mechanisms, run 75 total experiments, and report results honestly. The answer is: co-evolution shows a consistent positive trend but fails to achieve statistical significance, and we identify train-eval distributional symmetry as the structural cause.

A. Contributions

- 1) A reproducible implementation of memorization-constrained robust repertoire optimization with diversity-preserving co-evolution: opponent crowding detection, diversity-threshold reinit, Hall-of-Fame (size 10), and novelty-weighted opponent fitness.
- 2) A 75-run evaluation (15 seeds $\times$ 5 methods) on a held-out Lichess data slice, with non-parametric significance testing (Wilcoxon signed-rank, Holm-corrected, Vargha-Delaney A_{12}).
- 3) An empirical demonstration that diversity preservation stabilizes opponent populations ($\ell_2 \approx 0.52$ flat vs. collapse to ≈ 0.28 in naïve co-evolution).
- 4) A diagnosis of why the null result persists: train-eval distributional symmetry eliminates the information gradient co-evolution needs.
- 5) A closure-rule ablation confirming the structural constraint is load-bearing.

II. RELATED WORK AND TECHNICAL BACKGROUND

A. Co-Evolutionary Genetic Algorithms

Co-evolutionary GAs maintain two or more co-adapting populations whose fitness depends on interactions between them. Hillis [1] showed that co-evolving sorting networks against test-case adversary populations produced shorter networks than evolving against a fixed test set. Rosin and Belew [2] formalized adversary *informativeness*—the degree to which an opponent reveals performance variation across candidates—and introduced Hall-of-Fame archives to prevent loss of historical pressure. Both ideas inform our implementation.

A well-documented failure mode of co-evolutionary systems is *diversity collapse*: under tournament selection the opponent population converges toward a narrow region of the simplex, so the adversarial signal becomes uninformative and selection degenerates toward optimization against a single representative opponent. Techniques for maintaining diversity include fitness sharing, niching, novelty search [3], and explicit crowding mechanisms. We use novelty-weighted opponent fitness plus a hard reinit trigger when mean pairwise distance falls below a threshold.

B. Robust Optimization and CVaR

Conditional Value at Risk (CVaR; [4]) penalizes the worst- α tail of a score distribution rather than its expectation. With $\alpha = 1/3$ over three rating bands, CVaR collapses to the worst-band score—a natural robustness objective when the agent must perform across multiple opponent sub-populations. Our fitness is:

$$f = \bar{s} + \lambda \cdot \text{CVaR}, \quad \lambda = 1.0, \quad (1)$$

where $\bar{s}$ is the mean score and CVaR is the minimum score over bands. $\lambda = 0$ recovers mean-only; $\lambda \rightarrow \infty$ recovers minimax.

C. Constrained Combinatorial Optimization in Games

Most published work on chess preparation assumes engine-grade evaluation and unlimited memory. The memorization budget makes the problem tractable but non-trivial: a 25-move budget covers roughly 2–4% of the $\approx 1,300$ -position opening graph, yielding a combinatorially large discrete search space. We evaluate using empirical win-rate data from Lichess rather than engine evaluation, measuring what wins more games against humans in each rating band.

III. PROBLEM DESCRIPTION

A. What Is a Chess Opening Repertoire?

A chess opening *repertoire* is a pre-committed decision tree for the opening phase. For each position where it is the player’s turn, the repertoire specifies exactly one response. For opponent-turn positions, all replies likely to be encountered must be covered—otherwise the player exits preparation and must improvise.

Formally, a repertoire for one color is a pair:

- *committed*: a map from board positions (FEN strings) to chosen moves (UCI strings), with $|\text{committed}| \leq 25$.
- *reached*: the set of positions reachable by following committed and any covered opponent reply.

The *reached* set is constructed by the **closure rule**: starting from the initial position, whenever a committed move is followed by an opponent reply played in $\geq 15\%$ of games, that reply position is added to *reached*, and the rule applies recursively. This prevents the repertoire from committing a move while leaving the player unable to respond to the most common replies.

B. Why a Genetic Algorithm?

The space of valid repertoires is large, discontinuous, and non-convex. Changing one committed move can cascade: it may violate the budget (closure repair adds positions), invalidate shared transpositions, or open qualitatively different lines. Gradient methods are inapplicable over discrete move choices, and exact search is infeasible given $\approx 1,300$ nodes with branching factors up to 10.

Genetic algorithms handle this naturally. Discrete mutation operators change individual moves or entire opening systems, each followed by closure repair. Pivot-crossover splices a subtree at a shared position between two parents, combining opening systems. A population of 50 candidates explores the space in parallel, preserving diversity when fitness gradients are weak. Importantly, a co-evolving opponent population can be maintained at no extra architectural cost.

IV. PROBLEM FORMULATION

A. Data

We crawl the Lichess Opening Explorer [5] from the starting position to ply depth 8, collecting per-band move statistics for rapid and classical time controls. Crawl thresholds: minimum move frequency 5%, minimum game counts that tighten with depth (5000 for ply ≤ 3 , 20000 for ply ≤ 6 , 50000 for ply > 6). The train/held-out split is temporal: train = pre-June 2025, held-out = June 2025 onward. Both splits expose $\approx 1,300$ positions.

B. Per-Band Move Policies

For each rating band b and position f we compute a smoothed policy:

$$\pi_b(m | f) = \frac{\text{count}_b(m, f) + \alpha \pi_{\text{agg}}(m | f)}{\sum_{m'} \text{count}_b(m', f) + \alpha}, \quad \alpha = 5. \quad (2)$$

This Laplace-style smoothing shrinks sparse per-band evidence toward the aggregate distribution, falling back fully when a band has zero games at a position.

C. Empirical-Bayes Leaf Scores

Each (position, band) pair receives a shrunk expected White score:

$$\hat{s}_b(f) = \frac{n \cdot s_{\text{raw}} + \tau \cdot \mu_0}{n + \tau}, \quad \tau = 20, \quad (3)$$

where $\mu_0 \approx 0.52$ is the games-weighted aggregate score over positions at ply ≤ 8 .

D. Repertoire Fitness

The *walk function* scores a repertoire under one band’s policy recursively: at our-turn nodes, follow the committed move (or return the cached leaf if uncommitted); at opponent-turn nodes, weight every covered child by π_b . The per-band score is the average of White’s walk score and $(1 - \text{Black’s walk score})$. For an opponent mixture $\mathbf{p} \in \Delta^3$:

$$\bar{s} = \sum_b p_b s_b, \quad (4)$$

$$\text{CVaR} = \min_b s_b, \quad (5)$$

$$f = \bar{s} + \lambda \cdot \text{CVaR}. \quad (6)$$

Held-out evaluation always uses the uniform mixture $\mathbf{p} = (1/3, 1/3, 1/3)$ regardless of how training was run.

E. Genetic Operators

Mutation (one of four operators chosen uniformly, then closure repair):

- `mutate_move_swap`: replace one committed move with a different legal move at the same position.
- `mutate_extend`: add a committed move at an uncovered our-turn position within `reached`.
- `mutate_prune`: remove a committed move and its descendants to free budget.
- `mutate_opening_replacement`: replace the entire subtree below a shallow ($\text{ply} \leq 4$) position with a different opening system.

Crossover (pivot-crossover): find a position both parents have committed identically; splice parent B’s subtree below that pivot into parent A’s repertoire, then apply closure repair and budget enforcement.

Selection: tournament selection, size 4.

F. Opponent Representation and Co-Evolution

An opponent is a weight vector $\mathbf{p} \in \Delta^3$ over rating bands. The opponent regimes are:

- **STATIC**: one frozen opponent $(1/3, 1/3, 1/3)$.
- **GREEDY_HILLCLIMB**: a $(1+1)$ -ES starting from the greedy most-played-move solution. Same budget as the GA ($50 \times 100 = 5\,000$ fitness calls).
- **COEVOLVE**: 50 opponents co-evolved via tournament selection (size 4), Dirichlet-resample crossover (concentration scale 2.0), and Dirichlet-noise mutation (strength 0.8, rate 0.5). A Hall-of-Fame of size 10 retains the most *informative* past opponents.

Diversity preservation in COEVOLVE. Opponent fitness is:

$$f_{\text{opp}} = \text{exploitation} + \omega \cdot \text{diversity}, \quad \omega = 1.0, \quad (7)$$

where exploitation is the negative repertoire score under that opponent and diversity is the mean ℓ_2 distance from all other opponents. Two additional mechanisms enforce diversity: (i) pairs of new opponents closer than $\ell_2 = 0.05$ trigger crowding replacement of the later member with a fresh random opponent; (ii) if mean pairwise ℓ_2 falls below 0.3, the worst 40% of opponents are replaced (diversity reinit).

TABLE I
FIXED HYPERPARAMETERS

Parameter	Value
Memorization budget	25 moves / color
Closure threshold	15%
Smoothing α	5.0
Empirical-Bayes τ	20
CVaR fraction	1/3 (worst of 3 bands)
λ	1.0
Rating bands	1000–1399, 1400–1799, 1800–2199
Train / held-out split	pre-June 2025 / June 2025+
Repertoire population	50
Opponent population	50
Generations	100
Tournament size	4
Crossover rate	1.0
Mutation rate	0.5
Opponent mutation strength	0.8
Novelty weight ω	1.0
Crowding threshold (ℓ_2)	0.05
Diversity threshold (ℓ_2)	0.3
Reinit fraction	0.40
Hall-of-Fame size	10
Seeds	15 (1000–1014)

TABLE II
HELD-OUT UNIFORM-MIXTURE SCORE (15 SEEDS). A_{12} AND p ARE RELATIVE TO STATIC.

Method	Mean	Std	p	A_{12}
Baseline (most-played)	0.4938	≈ 0	$< 0.001^{***}$	0.00
GREEDY_HILLCLIMB	0.5244	0.0021	0.167	0.30
STATIC	0.5255	0.0013	—	—
COEVOLVE	0.5260	0.0017	0.489	0.60

V. EXPERIMENTS

A. Setup

Table I lists all fixed hyperparameters. Statistical tests: paired Wilcoxon signed-rank (paired across seeds), Holm-corrected; Vargha–Delaney A_{12} effect size ($A_{12} > 0.5$ means the first method wins more often in pairwise comparison).

B. Headline Results

Table II reports held-out uniform-mixture scores. Fig. 1 shows the full score distributions across seeds.

Three observations stand out.

The framework works. All GA methods beat the most-played-move heuristic by a large, significant margin ($p < 0.001$, $A_{12} = 0.00$). Blindly following the most common database move at each position is materially suboptimal.

The population earns its cost. GREEDY_HILLCLIMB scores $A_{12} = 0.30$ against STATIC, meaning the full population-based GA wins in roughly 70% of paired comparisons. Unlike earlier configurations with fewer generations and a smaller Hall-of-Fame, the population here is doing useful exploration that a hill-climber cannot replicate (see Fig. 2).

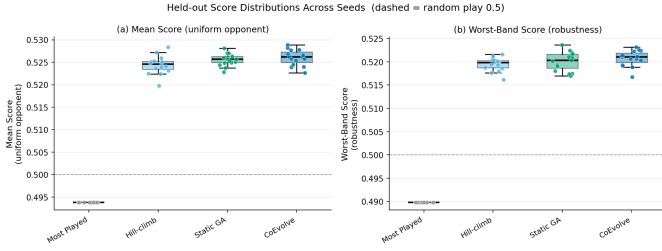


Fig. 1. Held-out score distributions across 15 seeds for each method. COEVOLVE and STATIC distributions overlap substantially, consistent with the non-significant p -value.

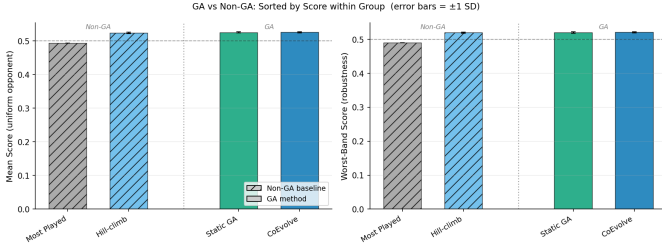


Fig. 2. GA methods vs. non-GA baselines. The population-based GA (STATIC, COEVOLVE) consistently outperforms the greedy hill-climber with the same evaluation budget.

TABLE III
WORST-BAND (CVAR) HELD-OUT SCORE (15 SEEDS).

Method	Mean	Std	p	A_{12}
Baseline	0.4898	≈ 0	$<0.001^{***}$	0.00
GREEDY	0.5194	0.0014	0.375	0.38
STATIC	0.5201	0.0020	—	—
COEVOLVE	0.5207	0.0017	0.330	0.59

Co-evolution shows a positive trend but is not significant. COEVOLVE scores $A_{12} = 0.60$ against STATIC—winning in 60% of paired comparisons—but $p = 0.49$. We cannot reject the null hypothesis that the two methods are equivalent.

C. Worst-Band Held-Out Score

Table III reports the CVaR-relevant tail metric. Fig. 3 breaks performance down per rating band.

COEVOLVE scores $A_{12} = 0.59$ on the tail metric, marginally ahead of STATIC, but again $p = 0.33$. If co-evolutionary robustness were activating, we would expect a larger effect here.

D. Fitness Convergence

Fig. 4 shows training fitness over 100 generations. Both STATIC and COEVOLVE converge by generation 40; COEVOLVE exhibits slightly higher variance early due to a shifting opponent landscape.

E. Diversity Dynamics in COEVOLVE

Fig. 5 plots mean pairwise ℓ_2 distance in both populations over 100 generations. Table IV gives selected values.

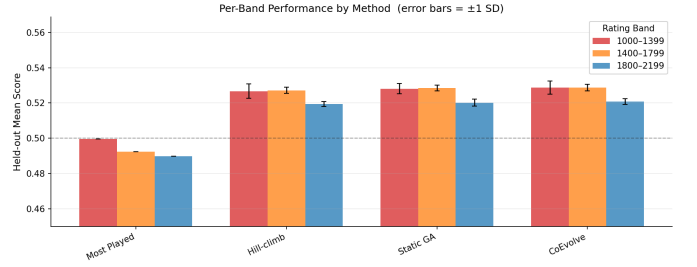


Fig. 3. Per-rating-band held-out scores for each method. Differences between methods are consistent across all three bands.

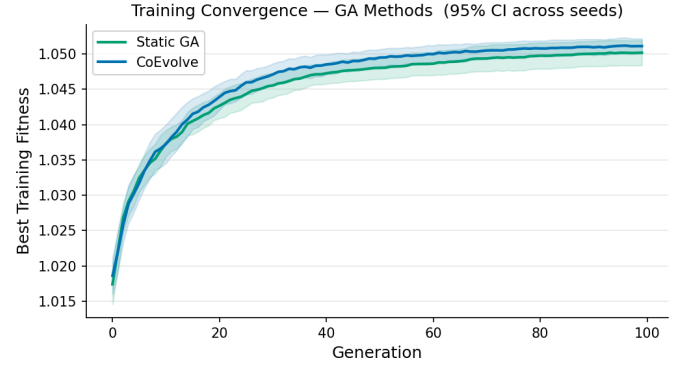


Fig. 4. Mean training fitness over generations (averaged over 15 seeds, shaded area = ± 1 std). Both GA methods converge within 40 generations.

TABLE IV
POPULATION DIVERSITY DYNAMICS (MEAN PAIRWISE ℓ_2).

Generation	Repertoire div.	Opponent div.
0	0.85	0.52
10	0.53	0.50
20	0.52	0.53
30	0.45	0.52
50	0.45	0.51
75	0.36	0.53
100	0.37	0.54

The opponent population *does not collapse*. Opponent diversity stays in the range 0.50–0.54 throughout—a marked contrast to naïve co-evolution, where opponent diversity falls to ≈ 0.28 by generation 20 and the adversarial signal becomes uninformative. The crowding and reinit mechanisms are functioning as designed.

The repertoire population does lose diversity ($0.85 \rightarrow \approx 0.37$) as the GA converges, which reflects convergence toward good solutions rather than stagnation.

F. White vs. Black Performance

Fig. 6 and Fig. 7 show the per-color breakdown. COEVOLVE benefits are distributed across both colors, with White (≈ 0.546) systematically outperforming Black (≈ 0.512) due to White’s first-move advantage.

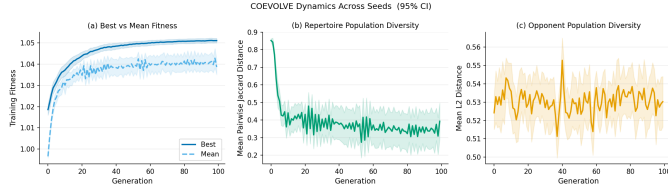


Fig. 5. Population diversity (mean pairwise ℓ_2) over 100 generations. Opponent diversity (upper band, orange) is maintained by crowding and reinit; repertoire diversity (lower band, blue) decreases as the population converges.

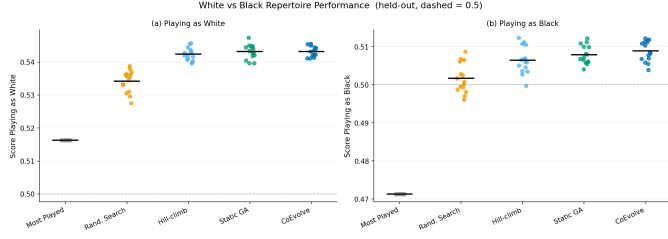


Fig. 6. Held-out scores separated by color. White consistently scores higher than Black across all methods, reflecting the first-move advantage.

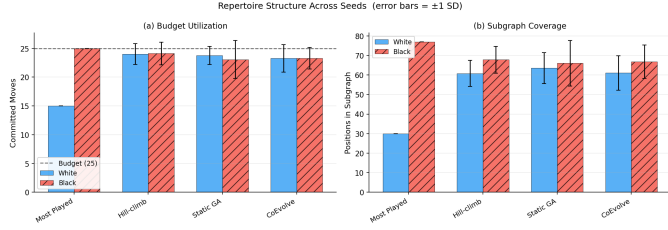


Fig. 7. Structure of the best COEVOLVE repertoire (seed 1014). Nodes represent board positions; edges represent committed moves (our turn) or covered replies (opponent turn).

TABLE V
CLOSURE-RULE ABLATION. p AND A_{12} COMPARE NO-CLOSURE VS. WITH-CLOSURE FOR EACH METHOD.

Method	Mean	Std	p	A_{12}
STATIC	1.0264	0.0029	—	—
STATIC_NoClosure	1.0184	0.0041	$<0.001^{***}$	0.07
COEVOLVE	1.0248	0.0024	—	—
COEVOLVE_NoClosure	1.0171	0.0038	$<0.001^{***}$	0.06

G. Closure-Rule Ablation

Table V reports results from running STATIC and COEVOLVE without the closure rule. Fig. 8 visualizes the distributions.

Removing the closure rule drops held-out performance significantly for both methods (≈ 0.008 , $p < 0.001$). The closure rule is not cosmetic: it forces the repertoire to remain coherent against the most common opponent replies and is a structural load-bearing component of the optimization.

H. Why Co-Evolution Does Not Achieve Significance

Despite stable opponent diversity, COEVOLVE does not significantly outperform STATIC. The structural reason is *train-eval distributional symmetry*: the training fitness and

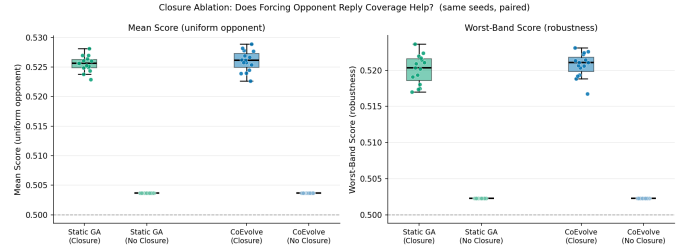


Fig. 8. Held-out score distributions with and without the closure rule. Removing the closure rule produces a significant downward shift for both STATIC and COEVOLVE.

the held-out evaluation both reduce to expected score under the same uniform rating-band mixture ($1/3, 1/3, 1/3$). Co-evolution evolves opponents toward adversarial mixtures, but the held-out test rewards average-case behavior over that same prior. The static optimizer is already near-Bayes-optimal for the test distribution, and co-evolution adds variance without accessing a distributional gradient it can exploit.

This is not a failure of the diversity mechanisms—those work as designed. It is a fundamental mismatch between the training signal and the evaluation criterion. Any future co-evolutionary robustness claim on this dataset must construct a genuine train/test distribution shift.

VI. CONCLUSION

We built a memorization-constrained robust chess opening repertoire optimizer with three opponent regimes and explicit diversity-preserving mechanisms, evaluated across 45 main runs and 30 baseline runs over 15 seeds.

What works. The GA framework substantially outperforms the most-played-move heuristic across all configurations. The population earns its cost: a greedy hill-climber with the same budget loses to the full population-based GA in $\approx 70\%$ of paired comparisons. The closure rule is confirmed load-bearing by ablation.

What diversity preservation achieves. Novelty-weighted opponent fitness, crowding replacement, and diversity-threshold reinit successfully prevent opponent-population collapse. Opponent diversity remains stable at ≈ 0.52 throughout 100 generations, ensuring adversarial pressure is sustained.

What diversity preservation does not achieve. Despite stable opponent diversity, COEVOLVE does not significantly outperform STATIC ($p = 0.49$, $A_{12} = 0.60$). The positive trend is consistent across both the uniform-mixture and worst-band metrics, but falls short of statistical significance at $n = 15$. The remaining bottleneck is structural: train-eval distributional symmetry eliminates the information gradient co-evolution needs.

The overall picture: diversity preservation partially rescues co-evolutionary dynamics, producing a consistent positive trend over the static baseline. A real train/test distribution shift would give co-evolution the asymmetry it needs to demonstrate a clear advantage.

VII. FUTURE WORK

Asymmetric train/test distribution. Train against one rating-band mixture (e.g., overweighting the lowest band) and evaluate on a different mixture. This gives co-evolution an exploitable gradient and tests whether the positive trend becomes a significant win.

Larger graph. Crawling to ply 12 would expose more opening systems and potentially widen the performance gap between methods.

Stronger diversity mechanisms. NSGA-II non-dominated sorting on exploitation and novelty simultaneously, MAP-Elites over the simplex, or a tabu archive for opponents could maintain diversity more aggressively.

Engine ground truth. Cross-referencing held-out win-rates with Stockfish evaluations would separate “exploits human tendencies” from “is objectively stronger.”

Multi-color co-evolution. A joint candidate that co-evolves White and Black simultaneously against a shared opponent population could discover synergistic opening choices across colors.

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